WEEK -4

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 13-02-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 4)

B. **Sprint / Issue Goal:** Design and validate the system's image processing workflow by creating a UML activity diagram based on reference research.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:**

1. UML Activity Diagram (Image Processing Workflow)

B. **Diagrams Revised:**

1. Ensured correct representation of parallel processing (Vegetation Analysis & Apple Detection).
2. Validated associations between:
   I. OrchardImage
   II. ImagePreprocessor
   III. HealthAnalysis
   IV. AppleDetection
   V. YieldEstimation
3. Improved responsibility allocation across processing modules.
4. Improved processing pipeline structure.
5. Refined Class Diagram after performing dynamic analysis.

C. **OO Concept Applied:**

A. **Inheritance:** User → Farmer, Admin

B. **Encapsulation:** Each processing class has defined responsibilities.

C. **Association:** Orchard Image linked to processing modules.

D. **Modular Design:** Independent classes for preprocessing, detection, analysis, and estimation.

E. **Parallelism Representation:** Fork and join nodes used in Activity Diagram.

Dynamic Analysis Performed:

Dynamic analysis was conducted to understand how the system reacts to events and user interactions during runtime.

The system behaviour is as follows:

1. Drone mission is triggered.
2. Orchard image is uploaded.
3. The system validates the uploaded image.
    - If invalid → Request re-upload.
    - If valid → Proceed to preprocessing.
4. Image preprocessing is performed.
5. After preprocessing, two operations execute in parallel:
    - Vegetation health analysis
    - Apple detection
6. Outputs from both processes are combined.
7. Yield estimation is calculated.
8. Health map and final report are generated.
9. The farmer views the results.

This dynamic modelling ensures the system correctly handles:

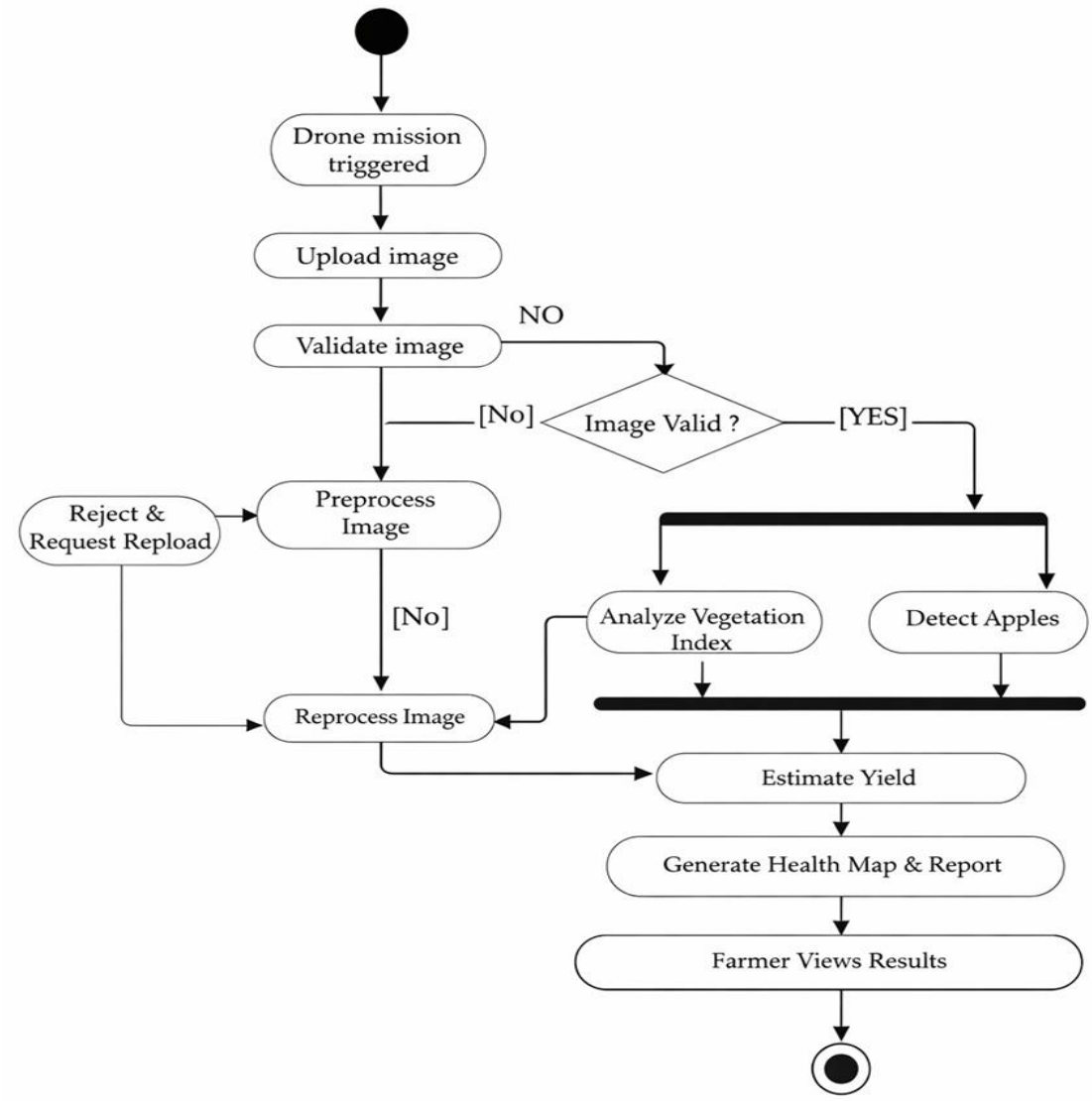
1. User-triggered events
2. Decision points
3. Failure handling
4. Parallel processing flows

Main Workflow of the System

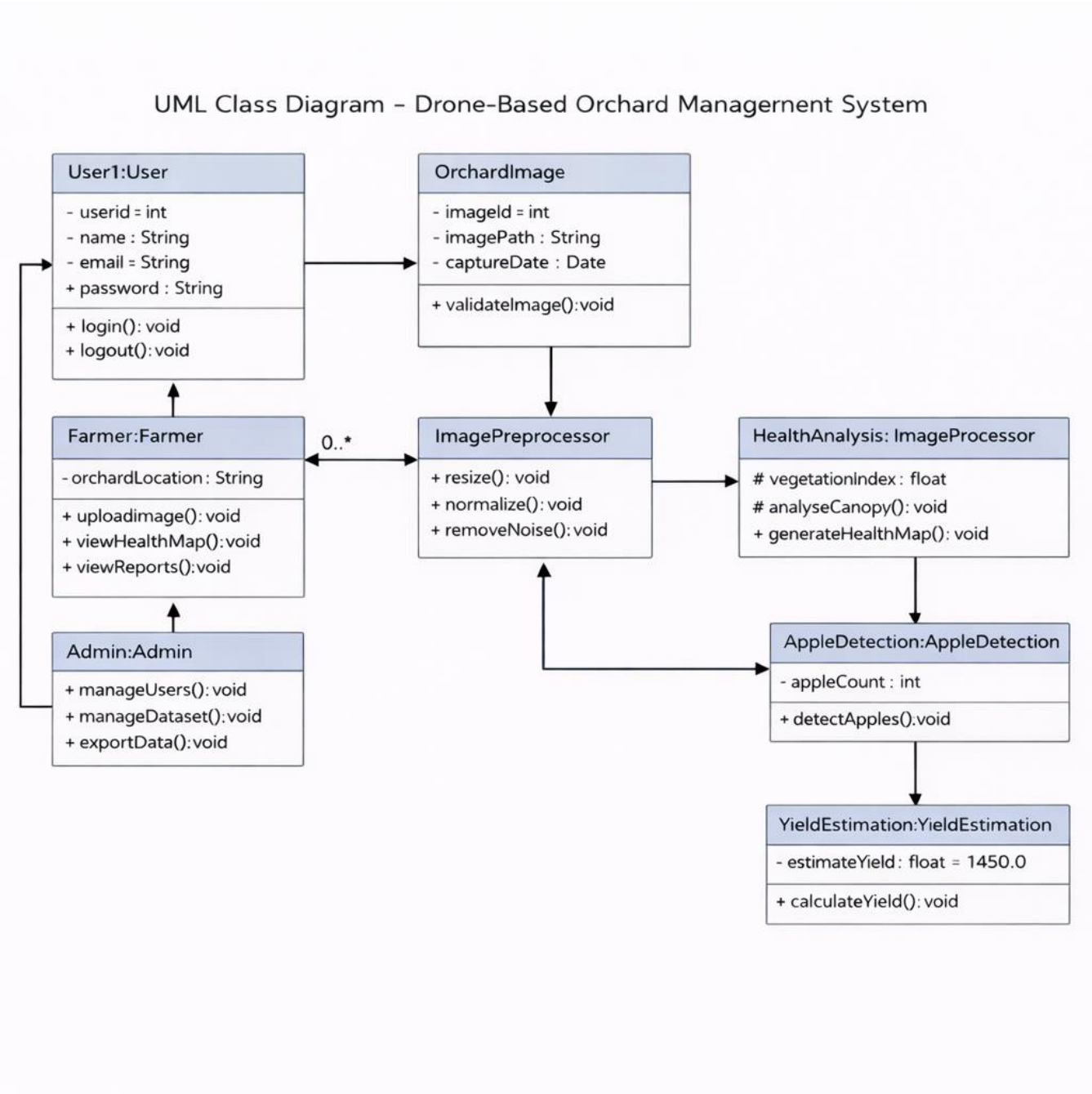
1. Drone mission triggered
2. Image upload
3. Image validation
    - If invalid → Reject & request re-upload
    - If valid → Continue
4. Image preprocessing
5. Parallel processing:
    - Analyse vegetation index
    - Detect apples

6. Combine analysis results
7. Estimate yield
8. Generate health map & report
9. Farmer views results

Activity Diagram



Updated class diagram



3. Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Perform dynamic analysis
2. Design UML Activity Diagram
3. Define image processing workflow
4. Validate workflow using reference research
5. Refine class diagram based on dynamic behaviour

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Active Branches: main
2. Documentation updated
3. UML diagrams uploaded

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:** Not applicable (design phase – no code yet)

4. Testing Summary

A. **Total Unit Tests:** Not applicable in design phase

B. **Pass Rate:** Not applicable in design phase

C. **Key Bug Identified:** No major issues were identified during this phase.

5. Goals for Next Week

A. Define the detailed logic of system interactions by creating a UML Sequence.

B. Collect the dataset for project.

Mentor Signature **Project Guide Signature**